

An Introduction to Differential Forms and Multi-Dimensional Stokes Theorem

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1 Introduction

In this paper, we will closely follow chapter 10 of [1], Integration of Differential Forms. We will begin with a proof of Stone-Weierstrass theorems, before introducing differential forms, their properties. We will define notions of differentiation and multiplication of differential forms, by way of examples, before concluding with Stokes Theorem and properties of closed forms.

2 Differential Forms and Properties

Definition 2.1 (*k*-surface). *Let $E \subset \mathbb{R}^n$ be open. A k -surface in E is a continuously differentiable map $\Phi : D \rightarrow E$, where D is a compact subset of $\mathbb{R}^k$.*

Note that we have defined a k -surface in terms of a mapping, not a set in $\mathbb{R}^k$. This is similar to how we view continuously differentiable curves in $\mathbb{R}^n$ (1-surfaces) as mappings from some subset of $\mathbb{R}^m$ for some m . This view of surfaces will allow us to integrate over them similarly to how we integrate over a curve.

Definition 2.2 (Differential Form). *Let $E \subset \mathbb{R}^n$ be open. A differential form of order $k \geq 1$, which we will call k -form, is a function*

$$\omega = \sum a_{i_1 \dots i_k}(x) dx_{i_1} \wedge \dots \wedge dx_{i_k},$$

The summation above will have $\binom{n}{k}k!$ terms, which we index by i . Each i term is associated to a permutation of k indices chosen from $\{1, \dots, n\}$.

For a given i -th term of the summation above, the term consists of a continuous function $a_{i_1 \dots i_k} : E \rightarrow \mathbb{R}$, which is paired with a wedge product term, whose indices $i_1, \dots, i_k$ are the preselected choice of k indices associated to this term of the summation.

ω gives each k -surface in $E \subset \mathbb{R}^n$ a real number, by way of establishing a notion of integration of ω on a surface Φ .

$$\omega(\Phi) = \int_{\Phi} \omega = \int_D \sum a_{i_1 \dots i_k}(\Phi(u)) \frac{\partial(x_{i_1}, \dots, x_{i_k})}{\partial(u_1, \dots, u_k)} du.$$

Here, again, the summation is over all choices of k distinct indices from $1, \dots, n$ and D is the parameter domain of Φ , which we integrate over using the iterated Riemann integral. Finally, $\frac{\partial(x_{i_1}, \dots, x_{i_k})}{\partial(u_1, \dots, u_k)}$ is the determinant of the $k \times k$ submatrix taken from the partial derivative matrix of the k -surface Φ that is given by the columns corresponding to $x_{i_1}, \dots, x_{i_k}$.

The wedge product (or cross product) $\wedge$ is an operator that is linear in both of its inputs and antisymmetric, meaning that $a \wedge b = -b \wedge a$. In the definition, the wedge products represent k -dimensional oriented volume.

Notice that integrating Φ over its parameter domain D matches our intuition for how we integrate over a curve $f(x) : [a, b] \rightarrow \mathbb{R}$, where we integrate over the parameter domain $[a, b]$.

Example 1: In general, in $\mathbb{R}^3$, a 2-form ω is of the form

$$\omega = a_{12}(x)dx_1 \wedge dx_2 + a_{13}(x)dx_1 \wedge dx_3 + a_{23}(x)dx_2 \wedge dx_3$$

Example 2: Let $\omega = xydy \wedge dz + xdz \wedge dx + 3zxdx \wedge dy$ and let S be the surface

$$\begin{aligned} x(u, v) &= u + v \\ y(u, v) &= u - v \\ z(u, v) &= uv \end{aligned}$$

The partial matrix of the surface S is then

$$\begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ v & u \end{bmatrix}$$

From this we get that

$$\begin{aligned} dx \wedge dy &= \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2 \\ dy \wedge dz &= \det \begin{pmatrix} 1 & -1 \\ v & u \end{pmatrix} = u + v \\ dz \wedge dx &= \det \begin{pmatrix} v & u \\ 1 & 1 \end{pmatrix} = v - u \end{aligned}$$

Now, by definition of a differential form, we have that

$$\begin{aligned} \int_S \omega &= \int_0^1 \int_0^1 ((u+v)(u-v)(u+v) + (u+v)(v-u) + 3uv(u+v)(-2))dudv \\ \int_S \omega &= \int_0^1 \int_0^1 (u^3 - 5u^2v - u^2 - 7v^2u + v^2 - v^3)dudv \\ \int_S \omega &= \int_0^1 ((\frac{1}{4}(1)^4 - \frac{5}{3}(1)^3v - \frac{1}{3}(1)^3 - \frac{7}{2}(1)^2v^2 + (1)v^2 - u(1)^3) - 0)dv \\ \int_S \omega &= \int_0^1 (-\frac{1}{12} - \frac{5}{6}v - \frac{5}{2}v^2 - v^3)dv = -2 \end{aligned}$$

Properties:

The integral of differential forms ω_1 and ω_2 over a k -surface Φ satisfies directly from the definition that $\forall c \in \mathbb{R}$ we have

- i) $\int_{\Phi} c\omega_1 = c \int_{\Phi} \omega_1$
- ii) $\int_{\Phi} \omega_1 + \omega_2 = \int_{\Phi} \omega_1 + \int_{\Phi} \omega_2$

We also have that $dx_i \wedge dx_i = 0 \forall i \in \{1, \dots, n\}$ since $dx_i \wedge dx_i = -dx_i \wedge dx_i$ by antisymmetry.

Definition 2.3 (Basic k -form). Any k -form of the form $dx_{i_1} \wedge \dots \wedge dx_{i_k}$ for some indices $i_1 \leq i_2 \leq \dots \leq i_k$ with each $i_j \in \{1, \dots, n\}$ is a basic k -form. We denote $dx_I = dx_{i_1} \wedge \dots \wedge dx_{i_k}$.

Definition 2.4 (Standard presentation). The standard presentation of a k -form ω expresses every element of the sum $\omega = \sum a_{i_1 \dots i_k}(x)dx_{i_1} \wedge \dots \wedge dx_{i_k}$ as a continuous function times a basic k -form. We denote

$$\omega = \sum_I b_I(x)dx_I$$

as the standard presentation of ω , where the summation is over all increasing choices of k indices from $\{1, \dots, n\}$.

Note that when converting a differential form ω into its standard presentation, the continuous functions $a_{i_1 \dots i_k}$ associated with each increasing index may be changed by a sign as a result of the antisymmetry of the wedge product.

3 Multiplication and Differentiation

Definition 3.1 (Multiplication of Differential Forms). *Let ω and λ be p - and q -forms, with standard presentations*

$$\omega = \sum_I b_I(x) dx_I, \lambda = \sum_J c_J(x) dx_J$$

, where I and J range over all increasing p - and q -indices taken from $\{1, \dots, n\}$ respectively. Their wedge product is

$$\omega \wedge \lambda = \sum_{I,J} b_I(x) c_J(x) dx_I \wedge dx_J$$

where I and J range independently over all their possible values just described above. For two fixed basic k -forms $I = dx_{i_1}, \dots, dx_{i_p}$ and $J = dx_{j_1}, \dots, dx_{j_q}$, their product $dx_I \wedge dx_J$, is

$$dx_{i_1} \wedge \dots \wedge dx_{i_p} \wedge dx_{j_1} \wedge \dots \wedge dx_{j_q}$$

The product of ω and λ in this definition can be seen as simply distribution. It treats ω and λ as if they were polynomials, with each term in ω or λ being a basic p -form or k -form respectively.

Definition 3.2 (Differentiation). *We begin with defining differentiation on a 0-form, which is just a continuously differentiable function f . Define*

$$df = \sum_{i=1}^n (D_i f)(x) dx_i$$

Extending this to a k -form ω with standard presentation $\omega = \sum b_I(x) dx_I$, we define

$$d\omega = \sum_I (db_I) \wedge dx_I$$

Example 3: Consider a 1-form $f = dx + xydy + zdz$ and a 0-form $g = 1 + y^2 + z$. We want to compute $df \wedge dg$. We will do this by computing partial derivatives of g , and wedging with the corresponding variable.

Notice that

$$dx \wedge D_x(g) = 0 \tag{3.0.1}$$

$$dy \wedge D_y(g) = 2ydy \tag{3.0.2}$$

$$dz \wedge D_z(g) = dz \tag{3.0.3}$$

This implies

$$dg = 2ydy + dz$$

Similarly we have

$$dx \wedge D_x(f) = ydx \wedge dy \tag{3.0.4}$$

$$dy \wedge D_y(f) = xdy \wedge dy = 0 \tag{3.0.5}$$

$$dz \wedge D_z(f) = dz \wedge dz = 0 \tag{3.0.6}$$

This implies

$$df = ydx \wedge dy$$

So then,

$$df \wedge dg = (ydx \wedge dy) \wedge (2ydy + dz) \tag{3.0.7}$$

$$= 2y^2 dx \wedge dy \wedge dy + ydx \wedge dy \wedge dz \tag{3.0.8}$$

$$= ydx \wedge dy \wedge dz \tag{3.0.9}$$

4 Change of Variables

Theorem 4.1 (Change of Variables).

$$\omega = \sum_I b_I(y) dy_I$$

Theorem 4.2. Suppose $E \subset \mathbb{R}^n$ open, and T is a $\mathcal{C}^1$ mapping of E into an open set $V \subset \mathbb{R}^m$. Let ω and λ be k - and m -forms, respectively. Then, we have that

$$i) (\omega + \lambda)_T = \omega_T + \lambda_T \text{ if } k = m$$

$$ii) (\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$$

$$iii) d(\omega_T) = (d\omega)_T \text{ if } \omega \text{ is } \mathcal{C}^1 \text{ and } T \text{ is } \mathcal{C}^2$$

Theorem 4.3. Suppose ω is a k -form in an open set $E \subset \mathbb{R}^n$, Φ is a k -surface in E , with parameter domain $D \subset \mathbb{R}^k$, and Δ is the k -surface in $\mathbb{R}^k$, with parameter domain D , defined by $\Delta(u) = u$ ($u \in D$). Then

$$\int_{\Phi} \omega = \int_{\Delta} \omega_{\Phi}$$

Theorem 4.4. Suppose T is a $\mathcal{C}^1$ -mapping of an open set $E \subset \mathbb{R}^n$ into an open set $V \subset \mathbb{R}^m$, Φ is a k -surface in E , and ω is a k -form in V . Then,

$$\int_{T\Phi} \omega = \int_{\Phi} \omega_T$$

5 Simplexes and Chains

Definition 5.1 (Standard simplex). Define the standard simplex Q^k to be the set of all $u \in \mathbb{R}^k$ that may be written as $u = \sum_{i=1}^k \alpha_i e_i$ such that $\alpha_i \geq 0 \forall i \in \{1, \dots, k\}$ and $\sum_{i=1}^k \alpha_i \leq 1$, where each e_i is a standard basis vector of $\mathbb{R}^k$.

Note that Q^k looks like a k -dimensional right triangle in the first quadrant, with its right angle point at the origin.

Definition 5.2 (Affine simplex). Recall that a map $f : X \rightarrow Y$ is affine if $f - f(0) : X \rightarrow Y$ is a linear transformation. Let $p_0, p_1, \dots, p_k$ be points in $\mathbb{R}^n$. We define the oriented affine k -simplex

$$\sigma = [p_0, p_1, \dots, p_k]$$

as a mapping. The mapping σ is a k -surface in $\mathbb{R}^n$ with parameter domain the standard k -simplex Q^k , which is defined on the standard basis of $\mathbb{R}^k$ as

$$\sigma(\alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_k e_k) = p_0 + \sum_{i=1}^k \alpha_i (p_i - p_0)$$

Importantly, $\sigma(0) = p_0$ and $\sigma(e_i) = p_i$ for each standard basis vector e_i . So, σ translates the standard simplex Q^k by p_0 so that its point at the origin is now at p_0 . Then, it linearly transforms each of the translated standard basis vectors e_i that define Q^k so that they fit the points $[p_1, p_2, \dots, p_k]$.

Definition 5.3 (Oriented Affine k -simplex). Let us assume that $p_0, p_1, \dots, p_k$ are points of $\mathbb{R}^n$. Let us then define σ to be the k -surface in $\mathbb{R}^n$ with parameter domain Q^k , given by the affine mapping

$$\sigma(\alpha_1 e_1 + \dots + \alpha_k e_k) = p_0 + \sum_{i=1}^k \alpha_i (p_i - p_0)$$

where α_i are as in the definition of standard simplex above. We then call σ the oriented affine k -simplex, and write

$$\sigma = [p_0, p_1, \dots, p_k]$$

Definition 5.4 (Affine Chain). Say we have an open set $E \subset \mathbb{R}^n$. An affine k -chain Γ on E is simply a collection of finitely many oriented affine k -simplexes $\sigma_1, \dots, \sigma_r$ in E . Note that it is not necessary for σ_i to be distinct, we may repeat the simplexes that occur in Γ . Furthermore, if we have a k -form ω in E , then we define

$$\int_{\Gamma} \omega = \sum_{i=1}^r \int_{\sigma_i} \omega.$$

To simplify our notation, we express Γ as

$$\Gamma = \sum_{i=1}^r \sigma_i.$$

Definition 5.5 (Boundary). For $k \geq 1$, given an oriented affine k -simplex, the boundary of this simplex is defined to be the affine $(k-1)$ -chain

$$\partial\sigma = \sum_{j=0}^k (-1)^j [p_0, \dots, p_{j-1}, p_{j+1}, \dots, p_k]$$

Definition 5.6 (k -chain). A finite collection Ψ of oriented k -simplexes $\Phi_1, \dots, \Phi_r$ of class $\mathcal{C}^2$ in V is called a k -chain of class $\mathcal{C}^2$ in V . If ω is a k -form in V , we define

$$\int_{\Psi} = \sum_{i=1}^r \int_{\Phi_i} \omega$$

Definition 5.7 (Boundary of oriented k -simplex). The boundary of the oriented k -simplex $\Phi = T \circ \sigma$ is defined to be the $(k-1)$ -chain

$$\partial\Phi = T(\partial\sigma)$$

6 Stokes' Theorem

Theorem 6.1 (Stokes' Theorem). If Ψ is a k -chain of class $\mathcal{C}^2$ in an open set $V \subset \mathbb{R}^m$ and if ω is a $(k-1)$ -form of class $\mathcal{C}^1$ in V , then

$$\int_{\Psi} d\omega = \int_{\partial\Psi} \omega$$

Proof. First note that is sufficient to show that the above holds for an oriented k -simplex, since, if this is true and we have a k -chain $\Psi = \sum \Phi_i$, then it is clear that

$$\int_{\Psi} d\omega = \sum \int_{\Phi_i} d\omega = \sum \int_{\partial\Phi_i} \omega = \int_{\partial\Psi} \omega$$

and the result holds. Thus, let us assume we have an oriented k -simplex Φ of class $\mathcal{C}^2$. Let

$$\sigma = [0, e_1, \dots, e_k]$$

be the oriented affine k -simplex with parameter domain Q^k , defined by the identity mapping. Since we know that Φ is also defined on Q^k , and Φ is $\mathcal{C}^2$, we know there exists an open set $E \subset \mathbb{R}^k$ such that $Q^k \subset E$. Moreover, we know there exists a $\mathcal{C}^2$ mapping, $T : E \rightarrow V$ such that $\Phi = T \circ \sigma$. By applying change of variables, and differentiation properties to the left hand side, we see

$$\int_{\Phi} d\omega = \int_{T\sigma} d\omega = \int_{\sigma} (d\omega)_T = \int_{\sigma} d(\omega_T)$$

Furthermore, now applying these properties to the right hand side, observe that

$$\int_{\partial\Phi} \omega = \int_{\partial(T\sigma)} \omega = \int_{T(\partial\sigma)} \omega = \int_{\partial\sigma} \omega_T$$

Putting these together, it remains to show

$$\int_{\sigma} d(\omega_T) = \int_{\partial\sigma} \omega_T.$$

Notice here that ω_T is a $(k-1)$ -form in E , if we show that

$$\int_{\sigma} d\lambda = \int_{\partial\sigma} \lambda$$

for any $(k-1)$ -form λ , and our simplex σ , then we are done!

Notice that if we have $k = 1$, a $(k-1)$ -form is then a 0-form ω , which is just a continuously differentiable function f . So then we have

$$\int_{\Phi} d\omega = \int_0^1 f'(u)du = f(1) - f(0) = \int_{\partial\Phi} \omega$$

by the Fundamental Theorem of Calculus and the definition of a differential form, so the $k = 1$ case is proved.

So, let us assume that $k > 1$. Let us then fix an integer r such that $1 \leq r \leq k$, and choose a function $f \in \mathcal{C}^1(E)$. Now, note that is simply enough to consider

$$= f(x)dx_1 \wedge \dots \wedge dx_{r-1} \wedge dx_{r+1} \wedge \dots \wedge dx_k$$

since any $(k-1)$ -form will be a sum of terms of the format above. If we let

$$\tau_i = [0, e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_k]$$

for $i=1, \dots, k$, then, by definition, the boundary of σ is given by

$$\partial\sigma = [e_1, \dots, e_k] + \sum_{i=1}^k (-1)^i \tau_i$$

Let us further define

$$\tau_0 = [e_r, e_1, \dots, e_{r-1}, e_{r+1}, \dots, e_k]$$

and notice that we can now express the boundary as

$$\partial\sigma = (-1)^{r-1} \tau_0 + \sum_{i=1}^k (-1)^i \tau_i$$

When considering these τ_i as "pieces" of our boundary, we can see each τ_i as a hyperplane spanned by each axis except the i -th axis. Then, we see τ_0 is the diagonal "face" that joins each of these τ_i to complete the boundary containing no points at the origin. We want to parameterize points that lie on these components of the boundary, so that we may simplify our integration. Notice that each τ_i has Q^{k-1} as a parameter domain. Well, say that we have $x = \tau_i(u)$ where $u \in Q^{k-1}$ for some $0 \leq i \leq k$. Then, we may consider the mapping defined by

$$(u_1, \dots, u_{r-1}, u_r, \dots, u_{k-1}) \rightarrow (x_1, \dots, x_{r-1}, x_{r+1}, \dots, x_k)$$

to be the mapping induced by τ_i .

Let J_i be the Jacobian of this mapping. Notice now, that when $i = 0$ and $i = r$, the mapping above is merely the identity mapping. This allows us to conclude that $J_0 = 1$ and $J_r = 1$. Notice, furthermore, that for any other value of i , by construction $x_i = 0$, which tells us that J_i has a row of zeroes. So, when $i = 0$ notice that we have the identity mapping, and when $i = r$, we also have the identity mapping as we remove exactly the vertex not in Q^{k-1} and then project with the identity into Q^{k-1} . So, unless $i = r$ or $i = 0$, $J_i = 0$. This yields that

$$\int_{\tau_i} \lambda = 0 \text{ for } i \neq 0 \text{ and } i \neq r$$

by definition of a differential form. So, then we have that $\int_{\partial\sigma} \lambda = (-1)^{r-1} \int_{\tau_0} \lambda + \int_{\tau_r} \lambda$ since we have the boundary expressed as

$$\partial\sigma = (-1)^{r-1} \tau_0 + \sum_{i=1}^k (-1)^i \tau_i.$$

Then since $J_i = J_0 = 1$, we get that

$$\int_{\partial\sigma} \lambda = (-1)^{r-1} \int (f(\tau_0(u)) - f(\tau_r(u)))du$$

And by definition of differentiation and antisymmetry we have

$$\begin{aligned} d\lambda &= (D_r f)(x)dx_r \wedge dx_1 \wedge \dots \wedge dx_{r-1} \wedge dx_{r+1} \wedge \dots \wedge dx_k \\ &= (-1)^{r-1} (D_r f)(x)dx_1 \wedge \dots \wedge dx_k \end{aligned}$$

So, then returning to the integral we have

$$\begin{aligned} \int_{\sigma} d\lambda &= (-1)^{r-1} \int_{Q^k} (D_r f)(x)dx. \\ \int_{\sigma} d\lambda &= \int_0^1 \int_0^1 \dots \int_0^1 \int_0^{1-(x_1+\dots+x_{r-1}+x_{r+1}+\dots+x_k)} (D_r f)(x)dx_r dx_k \dots dx_{r+1} dx_{r-1} \dots dx_1 \end{aligned}$$

Notice that this is equivalent to $\int_{\partial\sigma} \lambda$ as desired, so we are done. □

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References

- [1] W. RUDIN, *Principles of Mathematical Analysis*, McGraw-Hill, New York, 3rd ed., 1976.